CS4051

Information Retrieval

Week 03

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Spelling & Phonetic Corrections

Spelling Corrections

- Two principal uses
  - Correcting document(s) being indexed
  - Correcting user queries to retrieve “right” answers
- Two main flavors:
  - Isolated word
    - Check each word on its own for misspelling
    - Will not catch typos resulting in correctly spelled words
    - e.g., ***from → form***
  - Context-sensitive
    - Look at surrounding words,
    - e.g., ***I flew <u>form</u> Heathrow to Narita.***

Spelling Corrections

Return to Google's jobs pages

488941	britney spears	29	britent spears	9	brinttany spears	5	brney spears	3	britiy spears	2	brirreny spears
40134	brittany spears	29	brittnany spears	9	britanay spears	5	broitney spears	3	britmeny spears	2	brirtany spears
36315	brittney spears	29	britttany spears	9	britinany spears	5	brotny spears	3	britneeey spears	2	brirttany spears
24342	britany spears	29	btiney spears	9	britn spears	5	bruteny spears	3	britnehy spears	2	brirttney spears
7331	britny spears	26	birrtney spears	9	britnew spears	5	btiyney spears	3	britnely spears	2	britain spears
6633	briteny spears	26	breitney spears	9	britneyn spears	5	btrittney spears	3	britnesy spears	2	britane spears
2696	britteny spears	26	brinity spears	9	brittney spears	5	gritney spears	3	britnetty spears	2	britaneny spears
1807	briney spears	26	britenay spears	9	brtiny spears	5	spritney spears	3	britnex spears	2	britania spears
1635	brittny spears	26	britneyt spears	9	brtittney spears	4	bittny spears	3	britneyxxx spears	2	britann spears
1479	brintey spears	26	brittan spears	9	brtny spears	4	bnritney spears	3	britnity spears	2	britanna spears
1479	britanny spears	26	brittne spears	9	brytny spears	4	brandy spears	3	britntey spears	2	britannie spears
1338	britiny spears	26	btittany spears	9	rbitney spears	4	brbritney spears	3	britnyey spears	2	britannt spears
1211	britnet spears	24	beitney spears	8	birtiny spears	4	breatiny spears	3	britterny spears	2	britannu spears
1096	britiney spears	24	birteny spears	8	bithney spears	4	breetney spears	3	brittneey spears	2	britanyl spears
991	britaney spears	24	brightney spears	8	brattany spears	4	bretiney spears	3	brittnney spears	2	britanyt spears
991	britnay spears	24	brintiny spears	8	breitny spears	4	brfitney spears	3	brittnyey spears	2	briteeny spears
811	brithney spears	24	britanty spears	8	breteny spears	4	briattany spears	3	brityen spears	2	britenany spears
811	brtiney spears	24	britenny spears	8	brightny spears	4	brieteny spears	3	briytney spears	2	britenet spears
664	birtney spears	24	britini spears	8	brintay spears	4	briety spears	3	brltney spears	2	briteniy spears
664	brintney spears	24	britnwy spears	8	brinttey spears	4	britny spears	3	broteny spears	2	britenys spears
664	briteney spears	24	brittni spears	8	briotney spears	4	briittany spears	3	brtaney spears	2	britianey spears
601	bitney spears	24	brittnie spears	8	britanys spears	4	brinie spears	3	brtiiany spears	2	britin spears
601	brinty spears	21	biritney spears	8	britley spears	4	brinteney spears	3	brtinay spears	2	britinary spears
544	brittaney spears	21	birtany spears	8	britneyb spears	4	brintne spears	3	brtinney spears	2	britmy spears
544	brittnay spears	21	biteny spears	8	britnrey spears	4	britaby spears	3	brtitany spears	2	britnaney spears
364	britey spears	21	bratney spears	8	britnty spears	4	britaey spears	3	brtiteny spears	2	britnat spears
364	brittiny spears	21	britani spears	8	brittner spears	4	britainey spears	3	brtnet spears	2	britnbey spears
329	brtney spears	21	britanie spears	8	brottany spears	4	britinie spears	3	brytiny spears	2	britndy spears
269	bretney spears	21	briteany spears	7	baritney spears	4	britinney spears	3	btney spears	2	britneh spears
269	britneys spears	21	brittay spears	7	birntey spears	4	britmney spears	3	drittney spears	2	britneney spears
244	britne spears	21	brittinay spears	7	biteney spears	4	britnear spears	3	pretney spears	2	britney6 spears
244	brytney spears	21	brtany spears	7	bitiny spears	4	britnel spears	3	rbritney spears	2	britneye spears
220	breatney spears	21	brtiany spears	7	breateny spears	4	britneuy spears	2	barittany spears	2	britneyh spears
220	britiany spears	19	birney spears	7	brianty spears	4	britnewy spears	2	bbbritney spears	2	britneym spears
199	britnney spears	19	briirtney spears	7	brintye spears	4	britnmey spears	2	bbitney spears	2	britneyyy spears
163	britnry spears	19	britnaey spears	7	britianny spears	4	brittaby spears	2	bbritny spears	2	britnhey spears
147	breatny spears	19	britnee spears	7	britly spears	4	brittery spears	2	bbrittany spears	2	britnjey spears
147	brittiney spears	19	britony spears	7	britnej spears	4	britthey spears	2	beitany spears	2	britnne spears
147	britty spears	19	brittanty spears	7	britneyu spears	4	brittnaey spears	2	beitny spears	2	britnu spears

Document Correction

- Especially needed for OCR'ed documents
  - Correction algorithms are tuned for this: "rn" / "m"
  - Can use domain-specific knowledge
    - E.g., OCR can confuse O and D more often than it would confuse O and I (adjacent on the QWERTY keyboard, so more likely interchanged in typing).
- But also: web pages and even printed material has typos
- Goal: the dictionary contains fewer misspellings

Isolated Word Correction

- Fundamental premise – there is a lexicon from which the correct spellings come
- Two basic choices for this
  - A standard lexicon such as
    - Webster's English Dictionary
    - An "industry-specific" lexicon – hand-maintained
  - The lexicon of the indexed corpus
    - E.g., all words on the web
    - All names, acronyms etc.
    - (Including the mis-spellings)

Isolated Word Correction

- Given a lexicon and a character sequence Q, return the words in the lexicon closest to Q
- What's "closest"?
- We'll study several alternatives
  - Edit distance (Levenshtein distance)
  - Weighted edit distance
  - *n*-gram overlap

Edit Distance

- Given two strings S_1 and S_2, the minimum number of operations to convert one to the other
- Operations are typically character-level
  - Insert, Delete, Replace, (Transposition)
- E.g., the edit distance from ***dof*** to ***dog*** is 1
  - From ***cat*** to ***act*** is 2 (Just 1 with transpose.)
  - from ***cat*** to ***dog*** is 3.

Edit Distance – Levenshtein

```
EDITDISTANCE(s1, s2)
int m[i, j] = 0
for i ← 1 to |s1|
do m[i, 0] = i
for j ← 1 to |s2|
do m[0, j] = j
for i ← 1 to |s1|
do for j ← 1 to |s2|
    do m[i, j] = min{m[i − 1, j − 1] + if (s1[i] = s2[j]) then 0 else 1fi,
                     m[i − 1, j] + 1,
                     m[i, j − 1] + 1}
return m[|s1|, |s2|]
```

► **Figure 3.5** Dynamic programming algorithm for computing the edit distance between strings s_1 and s_2.

Edit Distance – Levenshtein

		f		a		s		t	
	0	1	1	2	2	3	3	4	4
c	1	*1*	*2*	2	3	3	4	4	5
	1	*2*	*1*	2	2	3	3	4	4
a	2	2	2	*1*	*3*	*3*	*4*	4	5
	2	3	2	*3*	*1*	*2*	*2*	3	3
t	3	3	3	3	2	*2*	*3*	*2*	*4*
	3	4	3	4	2	*3*	*2*	*3*	*2*
s	4	4	4	4	3	2	3	*3*	*3*
	4	5	4	5	3	4	2	*3*	*3*

► **Figure 3.6** Example Levenshtein distance computation. The 2×2 cell in the $[i, j]$ entry of the table shows the three numbers whose minimum yields the fourth. The cells in italics determine the edit distance in this example.

Using Edit Distance

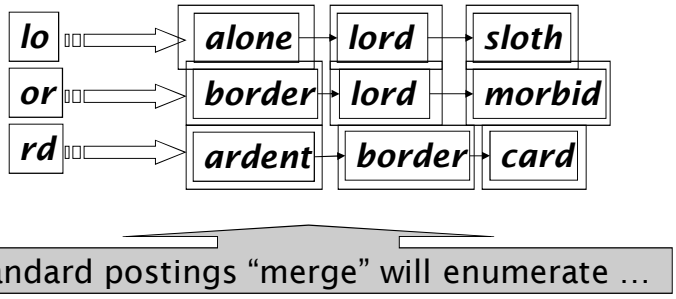
- Given query, first enumerate all character sequences within a preset (weighted) edit distance (e.g., 2)
- Intersect this set with list of “correct” words
- Show terms you found to user as suggestions
- Alternatively,
  - We can look up all possible corrections in our inverted index and return all docs … slow
  - We can run with a single most likely correction

n-gram Overlaps

- Enumerate all the *n*-grams in the query string as well as in the lexicon
- Use the *n*-gram index (recall wild-card search) to retrieve all lexicon terms matching any of the query *n*-grams
- Threshold by number of matching *n*-grams
  - Variants – weight by keyboard layout, etc.

2-grams for match

- Consider the query ***lord*** – we wish to identify words matching 2 of its 3 bigrams (***lo, or, rd***)



Context-Sensitive Spelling Corrections

- Text: ***I flew <u>from</u> Heathrow to Narita.***
- Consider the phrase query ***"flew <u>form</u> Heathrow"***
- We'd like to respond

  Did you mean "***flew from Heathrow***"?

because no docs matched the query phrase.

Context-Sensitive Spelling Corrections

- Need surrounding context to catch this.
- First idea: retrieve dictionary terms close (in weighted edit distance) to each query term
- Now try all possible resulting phrases with one word "fixed" at a time
  - ***flew from heathrow***
  - ***fled form heathrow***
  - ***flea form heathrow***
- **Hit-based spelling correction:** Suggest the alternative that has lots of hits.

Issues in Spelling Corrections

- We enumerate multiple alternatives for "Did you mean?"
- Need to figure out which to present to the user
- Use heuristics
  - The alternative hitting most docs
  - Query log analysis + tweaking
    - For especially popular, topical queries
- Spell-correction is computationally expensive
  - Avoid running routinely on every query?
  - Run only on queries that matched few docs

Soundex

- Class of heuristics to expand a query into phonetic equivalents
  - Language specific – mainly for names
  - E.g., ***chebyshev*** → ***tchebycheff***
- Invented for the U.S. census … in 1918

Soundex Algorithm

1. Retain the first letter of the word.
2. Change all occurrences of the following letters to '0' (zero):
   'A', E', 'I', 'O', 'U', 'H', 'W', 'Y'.
3. Change letters to digits as follows:
   - B, F, P, V → 1
   - C, G, J, K, Q, S, X, Z → 2
   - D,T → 3
   - L → 4
   - M, N → 5
   - R → 6

Soundex Algorithm

4. Remove all pairs of consecutive digits.
5. Remove all zeros from the resulting string.
6. Pad the resulting string with trailing zeros and return the first four positions, which will be of the form <uppercase letter> <digit> <digit> <digit>.

E.g., ***Herman*** becomes H655.

Soundex

- Soundex is the classic algorithm, provided by most databases (Oracle, Microsoft, …)
- How useful is soundex?
  - Not very – for information retrieval
- Zobel and Dart (1996) show that other algorithms for phonetic matching perform much better in the context of IR

Soundex Exercise

- Find two differently spelled proper nouns (different to the course example) whose soundex codes are the same and give their soundex code.
  - Mary, Nira (Soundex code = 5600).
- Find two phonetically similar proper nouns whose soundex codes are different.
  - Chebyshev, Tchebycheff
  - Rafi, Rafee